

Lab 1 – PatientPath Product Description

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1 Introduction

Although many hospitals and physician practices use digital tools to share patient information, exchanging records across organizations remains difficult. In a 2019 survey, 35% of office-based physicians used only fax, mail, or e-fax to share patient information with providers outside their organization (Pylypchuk et al., 2022). Hospitals have made progress with digital exchange. In 2023, 70% of hospitals engaged in all four exchange activities, which involve sending, receiving, finding, and integrating information, at least sometimes. However, only 43% did so routinely, and Figure 5 of the same report shows that just 17% sent summaries of care to most or all external long-term and post-acute care providers (Gabriel et al., 2024). Facilities that still rely on fax also face administrative work associated with sending and handling records through that process (Azar, 2020).

Patients moving between providers are the first to feel the effects of fragmented record exchange as many time-sensitive referrals and care coordination depend on timely exchange of medical information (Patel et al., 2018). These delays can lead to weakened specialist handoffs, referral loops that are never closed, and scheduling gaps that push back the next step in a patient's care (Patel et al., 2018). Patients may also end up repeating imaging or laboratory tests that were already completed at another facility, simply because the prior results did not arrive in time (De Swart et al., 2023). At the same time, medical providers and their staff take on additional administrative workload, spending time on manual record requests, resending faxes, and following up on information that should have already been transferred (Azar, 2020). These burdens are not felt equally across all facilities, as rural hospitals and lower-resourced organizations are less likely to engage in routine interoperability, meaning the patients who already have the fewest healthcare options often face the largest gaps in care (Gabriel et al., 2024).

While newer digital exchange systems have made progress, medical providers still face barriers when sharing records across organizations. These barriers include difficulty exchanging information with providers who use a different EHR vendor, locating providers' electronic addresses, and the additional costs of electronic exchange. Among office-based physicians who engaged in electronic exchange, 85% reported difficulty exchanging information with providers using a different EHR developer, and 71% reported that providers in their referral networks lacked the ability to exchange information electronically (Pylypchuk et al., 2022). Closing these gaps requires a way for organizations to share records across different systems while protecting patient information and keeping the process manageable for medical staff.

PatientPath is being developed to address these gaps in medical record transfer by providing a centralized, web-based platform for sharing records and managing referrals across organizations that use different EHR systems. Facilities would be able to continue using their existing software while authorized providers and staff use PatientPath to prepare transfers, make selected records available to another organization, and follow a referral's progress. The proposed platform would also include access controls and limited offline work for facilities that experience temporary internet outages. These features are intended to reduce repeated requests for missing records and make it easier for staff at both facilities to coordinate the next step in a patient's care.

2. PatientPath Product Description

PatientPath is a centralized, web-based medical record sharing and referral platform designed to connect medical facilities that use different electronic health record (EHR) systems. By bringing record transfers, referral tracking, and limited offline access together, the platform is intended to give medical providers and their staff a more consistent way to coordinate care across organizations. Facilities would be able to share information through PatientPath while continuing to use the EHR systems they already have. Its main goal is to reduce the delays caused by incomplete record transfers and repeated follow-up requests.

The main objectives of PatientPath are to shorten the time it takes to send medical records, reduce repeated requests for missing information, and make referral status easier for authorized staff to follow. In addition to improving this process, the platform is designed to protect patient information and support facilities that use different EHR products or experience temporary internet outages. As more healthcare organizations begin using the platform, the design should allow it to grow with their needs. PatientPath remains focused on sharing records and coordinating referrals, so it will not provide medical diagnoses or treatment recommendations, manage hospital billing or appointment scheduling, or serve as an emergency communication system.

2.1 Key Product Features and Capabilities

One of the main features of PatientPath is secure digital medical record exchange, which is intended to give providers a more direct way to send information to another facility. An authorized provider would select or upload the records needed for a referral and make them available to the receiving organization. Once the transfer is complete, the receiving provider could retrieve the records through the platform without waiting for printed documents or fax transmissions. Checks for required information and documents would help staff notice missing items before sending a transfer, reducing the need to resend the same records. Providers would still need to review the records for clinical completeness and accuracy.

PatientPath is also designed to connect different EHR systems through application programming interfaces (APIs), using REST interfaces, Fast Healthcare Interoperability Resources (FHIR) R4, and other Health Level Seven (HL7) standards. FHIR provides a common structure for sharing electronic healthcare information and can be used alongside other HL7 standards (HL7, 2023). Separate connectors are planned for different EHR products so facilities can share records through PatientPath while continuing to use their existing software. These planned connections include Epic, Oracle Health, and eClinicalWorks, although each connection would depend on the interfaces available and would need to be tested before use.

In addition to transferring records, PatientPath is designed to help medical staff keep track of referrals as they move between organizations. Providers would be able to create a referral, choose the records to include, and give the receiving organization access from the same interface. Staff could then see whether the referral has been sent, received, or returned because more information is needed, while notifications would let them know when they need to take

action. The goal is to reduce the time staff spend calling another office or sending follow-up requests just to find out what happened to a referral.

Protecting patient information is another part of PatientPath's design, as medical records need to remain accessible to the right people while being protected from unauthorized access. Planned security features include encryption during transfer and storage, multi-factor authentication, and role-based access control. Permissions would limit who can share records within an organization, while audit logs would keep a record of who accessed or transferred information. These features support the HIPAA Security Rule's safeguards for access control, authentication, audit controls, and transmission security (U.S. Department of Health and Human Services, 2024). The development plan also includes legal review and automated security checks as code is integrated and deployed. These technical features would support compliance efforts, while medical organizations would still need the required policies, staff training, and agreements (U.S. Department of Health and Human Services, 2024).

PatientPath's planned offline synchronization feature is intended to help facilities that experience unreliable internet access, including rural clinics. An encrypted cache on the user's device would keep previously available information accessible so authorized staff can continue limited work and prepare referrals during an outage. Once the connection returns, pending changes would synchronize with the platform. New transfers to another organization and updated referral statuses would still require an internet connection, so the feature would support work during an interruption without treating saved information as a current update.

2.2 Major Components (Hardware/Software)

PatientPath's software architecture is organized around the frontend, backend, and data storage, as shown in the Major Functional Component Diagram (MFCD). The local development environment uses React for the frontend, Express running on Node.js for the backend, and PostgreSQL running in Docker for the database. Prisma provides the tools used by the backend to access the database and apply changes to its structure.

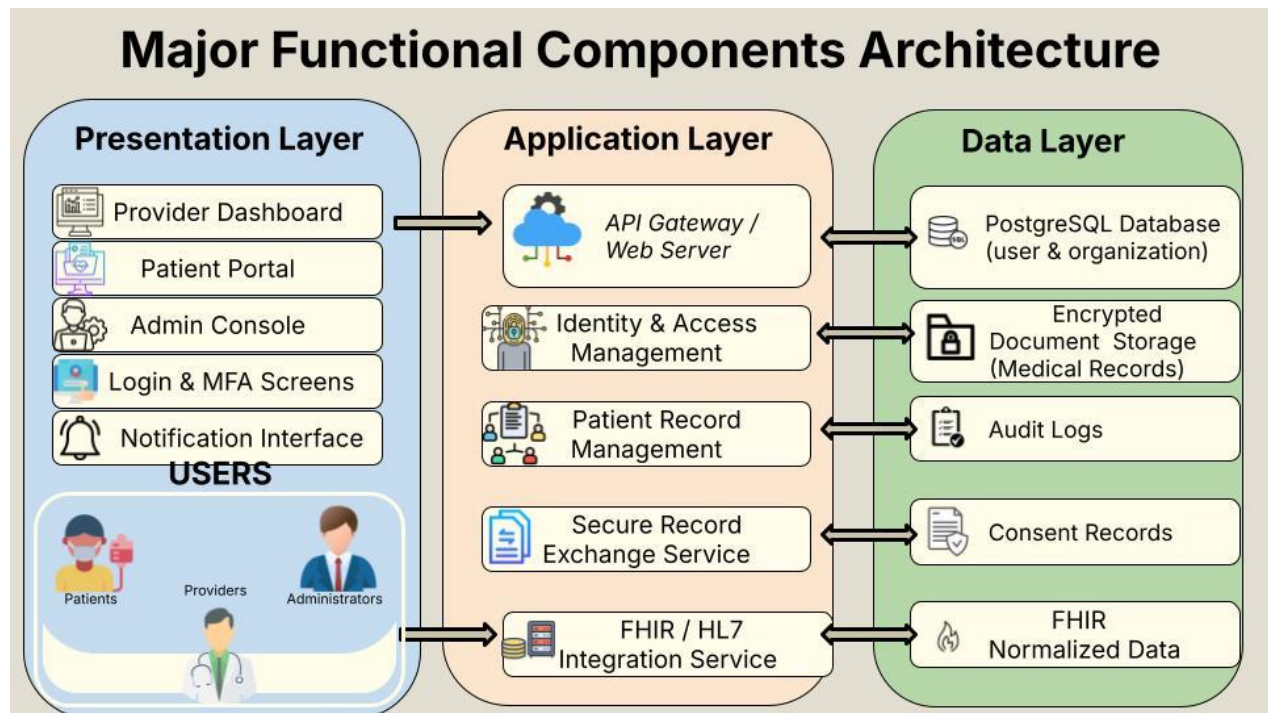


Figure 1: Major Functional Components Diagram

The React frontend provides the browser interface through which users interact with PatientPath. The full product design includes a provider dashboard for records and referrals, a patient portal for viewing available records and managing consent, and an administrative console for accounts and permissions. These views would keep the functions each person needs together while supporting their different roles. In the local setup, the frontend sends requests to the Express API instead of connecting directly to the database.

The backend uses Express on Node.js to receive requests from the frontend and communicate with the database. The local development setup includes API routes and permission checks, with temporary authentication to support testing. The full product design includes record exchange, referral status updates, notifications, and production authentication in this application layer. Planned REST interfaces and FHIR R4 and HL7 connectors would support exchange with other EHR systems. These connections are part of the proposed product and would require integration testing before use with Epic, Oracle Health, eClinicalWorks, or another EHR product.

PostgreSQL provides the application's relational database, while Prisma is used to generate the database client and apply changes to the database structure through migrations. The setup process also adds roles, permissions, and test accounts, giving the team a consistent starting point for development. The proposed data layer includes user and organization information, referrals, consent records, and access permissions. Encrypted storage for medical record files and audit logs are also part of the full product design. Access to these data sources would pass through the backend so permissions can be checked before information is returned.

For end users, the hardware requirements include a computer or tablet capable of running a supported web browser and a network connection for live record exchange. The planned offline feature would also require local storage for the encrypted cache. During development, each team member runs the frontend, backend, and database on their own computer. The team's Windows setup requires a computer that can run Docker Desktop alongside the browser and Node.js, with storage for the project files and local database. Hosting the full product would require server processing capacity, memory, network access, and storage for the application, database, medical records, and backups. Capacity would need to be sized for the expected number of users and volume of records.

The project's development tools include Node.js 22.18 or newer, npm for installing packages and running project commands, Docker Desktop for the local database container, and Git and GitHub for sharing code changes. PowerShell scripts automate setup, starting and stopping the application, and preparing updates, including Prisma migrations and test data. These tools support the local development environment, while production security, external EHR connections, and offline synchronization are requirements for the full product.

3. Identification of Case Study

PatientPath is being developed for hospitals, clinics, private practices, and specialist offices that need to share patient information with other medical facilities. Medical providers need access to records collected outside their own organizations, while administrative staff need a reliable way to send those records and follow up on referrals. Patients are also part of the intended user group, as the patient portal is designed to let them view available records and manage their consent. Improving coordination between these groups is a central goal of the platform, as delays can affect whether the next step in a patient's care is completed. Patel et al. (2018) found that longer wait times were associated with lower documented appointment completion rates, supporting the need for timely referrals.

The case study focuses on a rural clinic with limited information technology resources that refers patients to a regional hospital using a different EHR system. In this setting, the clinic's staff must deal with disconnected records and manual follow-up while also working around interruptions in internet access. This situation reflects the problems PatientPath is designed to address, since a referral depends on both facilities having access to the information needed for the patient's care. Research on oncological referrals in the Netherlands found that incomplete information transfers led to unnecessary repeat testing for some patients (De Swart et al., 2023). Although that study involved a different clinical setting, it helps explain why records already collected at one facility need to be available to the next provider.

A hypothetical example follows Dr. Sarah Chen, a primary care physician at a mid-sized rural clinic in Virginia, and her patient, Robert Mills, a 67-year-old man who needs specialist care for a newly diagnosed cardiac condition. Dr. Chen refers Robert to Dr. James Park, a cardiologist at a regional hospital 45 miles away, while Linda Torres, the clinic's administrative coordinator, handles the referral paperwork. Under the current process in this scenario, Linda faxes more than 15 pages of records to the hospital, only to receive a request for missing pages

two to three days later. She then has to send the records again, pushing Robert's appointment back by a week because the hospital is still waiting for the information.

With PatientPath, Dr. Chen could select the records needed for Robert's referral and give Dr. Park's organization access through the platform. Dr. Park's team would receive a notification when the records are available, while Linda could follow the referral's status through the administrative console without needing access to the clinical information. If the clinic loses its internet connection, the offline cache would allow Dr. Chen to continue limited work until the connection is restored and changes can synchronize. These features are intended to reduce the delays caused by missing records and repeated transfer attempts, helping Robert move to the next step in his care. The expected improvements would need to be confirmed through testing, and the receiving office would still be responsible for scheduling his appointment.

4. Glossary

Administrative console. The planned PatientPath interface for staff who manage accounts, permissions, and referral follow-up.

API. Application programming interface, which allows software systems to exchange information.

CI/CD. Continuous integration and continuous deployment, processes used to test and release software changes.

EHR. Electronic health record, a digital system used to store and manage patient information.

FHIR. Fast Healthcare Interoperability Resources, an HL7 standard for exchanging healthcare information.

FQHC. Federally qualified health center, a type of community health center.

HIE. Health information exchange, the electronic sharing of patient information between healthcare organizations.

HIPAA. Health Insurance Portability and Accountability Act, a U.S. law that includes requirements concerning the privacy and security of health information.

HL7. Health Level Seven, an organization that develops standards for exchanging healthcare information.

MFA. Multi-factor authentication, a sign-in method that requires more than one form of verification.

Offline cache. Encrypted information saved on a user's device for limited access during an internet outage.

Patient portal. The planned PatientPath interface where patients could view available records and manage consent.

PCP. Primary care provider, a clinician who provides a patient's general medical care.

RBAC. Role-based access control, a method of limiting access according to a user's assigned role.

Record transfer. The process of making selected patient records available to an authorized receiving organization through PatientPath.

Referral status. The stage of a PatientPath referral, such as sent, received, or returned for more information.

REST. Representational state transfer, an approach used to design interfaces that let applications communicate over the web.

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